1. Logarithmic & Exponential Functions

$$y = a^x \Leftrightarrow \log_a y = x$$

$$\log_a 1 = 0 \qquad \log_a a = 1$$

$$\log_a b^n \equiv n \log_a b$$

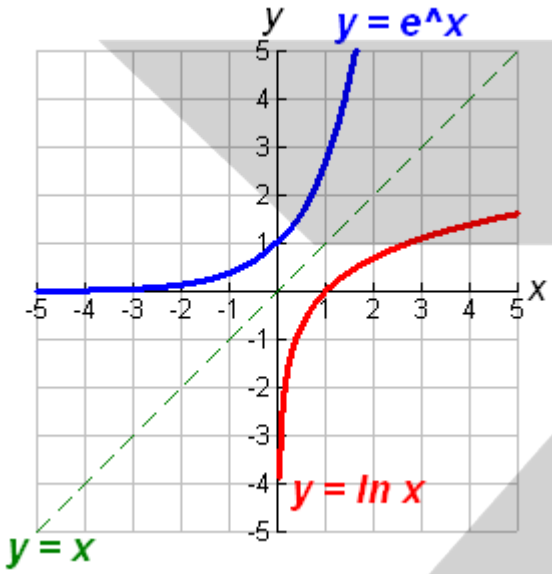
$$\log_a b + \log_a c \equiv \log_a bc$$

$$\log_a b - \log_a c \equiv \log_a \frac{b}{c}$$

$$\log_a b \equiv \frac{\log b}{\log a}$$

$$\log_a b \equiv \frac{1}{\log_b a}$$

1.2. Graphs of ln(x) and e^x



Copyright © 2025 ZNotes Education & Foundation. All Rights Reserved.
This document is authorised for personal use only by at Enderun Colleges on 05/04/26.